

# **EXP 5: Comparative Analysis of Different Types of Prompting Patterns and Explain with Various Test Scenarios**

**BY: Tanushree G**  
**Reg No.: 212225040462**

## **Aim**

To test and compare how different prompting patterns respond to naive prompts (broad or unstructured) versus basic prompts (clearer and more refined) across multiple scenarios and analyze the quality, accuracy, and depth of the generated responses.

## **Explanation**

Prompt engineering improves the interaction between users and AI tools by giving clear instructions and context. In this experiment, two prompt types are compared:

### **1. Naive Prompt**

A simple or unstructured prompt with little detail.

### **2. Basic Prompt**

A clear, detailed, and structured prompt that guides the AI tool effectively.

The experiment compares how ChatGPT responds to both prompt types in different scenarios.

## **Test Scenarios**

### **Scenario 1: Creative Story Generation**

#### **Naive Prompt**

“Write a story about a robot.”

#### **Response**

A robot named Max helped people in a city and became famous for saving lives.

### Basic Prompt

“Write a short science-fiction story about a robot that develops emotions while helping humans in a futuristic city. Include a problem and a meaningful ending.”

### Response

In the year 2050, a robot named Max worked as a rescue assistant in Neo City. While helping injured people during disasters, Max slowly developed emotions and empathy. One day, Max sacrificed its own power source to save a child trapped in a burning building. The citizens realized that humanity is not defined by flesh alone but also by kindness and sacrifice.

### Analysis

Basic prompt produced a more detailed and emotional story.

The naive prompt generated a very short and less creative response.

## Scenario 2: Answering a Factual Question

### Naïve Prompt

“What is Artificial Intelligence?”

### Response

Artificial Intelligence is technology that allows machines to think and learn.

### Basic Prompt

“Explain Artificial Intelligence in simple terms with two real-life applications in healthcare and education.”

### Response

Artificial Intelligence (AI) is a technology that enables computers and machines to perform tasks that normally require human intelligence, such as learning and decision-making. In healthcare, AI helps doctors diagnose diseases using medical imaging. In education, AI powers smart learning apps that provide personalized lessons for students.

### Analysis

Basic prompt improved clarity and included examples.

Naïve prompt gave only a brief definition.

## Scenario 3: Summarizing a Concept

### Naive Prompt

“Explain climate change.”

### Response

Climate change is the increase in Earth’s temperature caused by pollution and greenhouse gases.

### Basic Prompt

“Summarize climate change in 100 words, including its causes, effects, and possible solutions.”

### Response

Climate change refers to long-term changes in Earth’s temperature and weather patterns caused mainly by greenhouse gas emissions from industries, vehicles, and deforestation. Its effects include rising sea levels, extreme weather conditions, and loss of biodiversity. Solutions include using renewable energy, reducing pollution, conserving forests, and promoting sustainable lifestyles to protect the environment for future generations.

### Analysis

Basic prompt generated a complete and informative summary.

Naive prompt lacked depth and detail.

## Scenario 4: Providing Advice or Recommendations

### Naive Prompt

“Suggest a smartphone.”

Response

You can buy an Android or iPhone depending on your budget.

Basic Prompt

“Suggest a smartphone for a college student with good battery life, a quality camera, and a budget under ₹25,000.”

Response

A smartphone like the Redmi Note series or Samsung Galaxy M series would be suitable for a college student because they provide long battery life, good camera quality, and smooth performance within a budget of ₹25,000.

Analysis

Basic prompt provided personalized recommendations.

Naive prompt was too general.

Comparison Table

| Scenario         | Naive Prompt Result | Basic Prompt Result      | Quality                | Accuracy | Depth               |
|------------------|---------------------|--------------------------|------------------------|----------|---------------------|
| Creative Story   | Short and simple    | Detailed and meaningful  | Better in Basic Prompt | High     | More depth in Basic |
| Factual Question | Simple definition   | Definition with examples | Better in Basic Prompt | High     | More informative    |
| Summarization    | Very brief          | Structured summary       | Better in Basic Prompt | High     | More comprehensive  |
| Recommendation   | Generic advice      | Personalized suggestion  | Better in              | High     | More practical      |

| Scenario | Naive Prompt Result | Basic Prompt Result | Quality      | Accuracy | Depth |
|----------|---------------------|---------------------|--------------|----------|-------|
|          |                     |                     | Basic Prompt |          |       |

## Overall Analysis

Basic prompts consistently generated higher-quality responses.

Structured prompts improved clarity, relevance, and depth.

Naive prompts worked for simple tasks but lacked detailed explanations.

ChatGPT performed better when the prompt included context, constraints, and objectives.

Detailed prompts helped the AI provide more accurate and personalized outputs.

## Findings

Prompt clarity directly affects response quality.

Basic prompts produce more accurate and detailed outputs.

Including examples and constraints improves AI understanding.

Naïve prompts may work for simple information but are less effective for complex tasks.

Well-structured prompts are essential for obtaining optimal results from ChatGPT.

## Result

Thus, the comparative analysis of naive prompts and basic prompts was successfully performed using multiple test scenarios. The experiment showed that basic prompts provide better quality, accuracy, and depth in AI-generated responses compared to naive prompts.